

Online Low-Power Time-Varying IQ skew Estimation Scheme for Ultra-High-Speed Coherent Receivers

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Abstract We propose to use an online low-power side-path DSP to estimate the IQ skew of ultra-high-speed coherent receivers. Its effectiveness is experimentally verified in a 90 GBaud transmission experiment, with an estimation error of less than 1% of the symbol period. ©2026 The Author(s)

Introduction

The explosive growth of data traffic in optical networks has driven the urgent need for higher data rates and larger transmission capacities in coherent optical communication systems. With data rates in next-generation coherent systems reaching 1.6 Tbps, signals become extremely sensitive to receiver (Rx) in-phase/quadrature (IQ) skew due to high baudrate and modulation format [1]. While Rx IQ skew is typically calibrated during module production, calibration inaccuracies and temperature-induced variations during operation can result in time-varying IQ skew, significantly degrading signal quality. Consequently, online time-varying IQ skew estimation and compensation are critical for ultra-high-speed coherent receivers.

For the online estimation of time-varying Rx IQ skew, IQ-decoupled multiple-input multiple-output (MIMO) equalizers have been widely adopted in the main digital signal processing (DSP) chain and have proven effective [2-4]. However, compared to traditional 2×2 MIMO equalizers, this approach incurs higher power consumption for frequency-domain equalization. The Godard phase detector (GPD) is applied to the IQ branches to estimate IQ skew by comparing their optimal sampling phases [5-7]. However, the GPD is sensitive to bandwidth limitations and residual chromatic dispersion (CD), which degrades the estimation accuracy. Another approach involves transmitting specially designed symbol sequences. At the receiver, IQ skew is estimated based on the known training sequence [8]. However, this method requires dedicated sequences that are incompatible with standard frame structures, leading to increased system overhead and reduced spectral efficiency.

In this paper, we propose a low-power time-varying Rx IQ skew estimation scheme. Without modifying the main DSP architecture, we equalize the IQ branches in a side path to enable online IQ skew estimation using the equalizer coefficients. By sharing the main

path's chromatic dispersion compensation (CDC) coefficients, equalization coefficients and carrier phase recovery (CPR) coefficients with the side path, the algorithm achieves stable convergence and improved robustness against channel impairments. This approach can also reduce the computational frequency of the side path, thereby lowering the power consumption. We validate the proposed scheme in a 90 GBaud coherent transmission system with negligible estimation error and required optical signal-to-noise ratio (ROSNR) penalty.

Principle of the proposed scheme

We propose to use a side DSP to estimate the time-varying Rx IQ skew, as shown in Fig. 1(a). In the main-path DSP, the signal is processed by Rx IQ skew compensation, IQ imbalance compensation, fast Fourier transform (FFT), CDC, timing recovery, equalization, inverse fast Fourier transform (IFFT), CPR, symbol decision for mitigating channel impairments and recovering transmitted symbols. In the side-path DSP, the frequency-domain signal is extracted and IQ decomposition is performed as follows:

$$\begin{cases} XI(k) = \frac{1}{2} [X(k) + X^*(N-k)] \\ XQ(k) = \frac{1}{2j} [X(k) - X^*(N-k)] \end{cases} \quad (1)$$

where N is the FFT size and $k = 0, 1, \dots, N-1$. $X(k)$ represents the frequency-domain signal of the X-polarization complex signal $x_i + j \cdot x_q$, with $XI(k)$ and $XQ(k)$ being the frequency-domain signals of x_i and x_q , respectively. The same applies to Y polarization. The IQ branches undergo separate CDC after IQ decomposition, with CDC coefficients consistent with those in the main-path DSP. Subsequently, the side-path DSP performs time-varying Rx IQ skew estimation on IQ branches, as shown in Fig. 1(b).

The skew estimation scheme primarily comprises two 2×1 multiple-input single-output (MISO) equalizers, one 2×2 MIMO, an IFFT, a CPR and a symbol decision module. The two 2×1 MISOs are performed on the IQ branches of

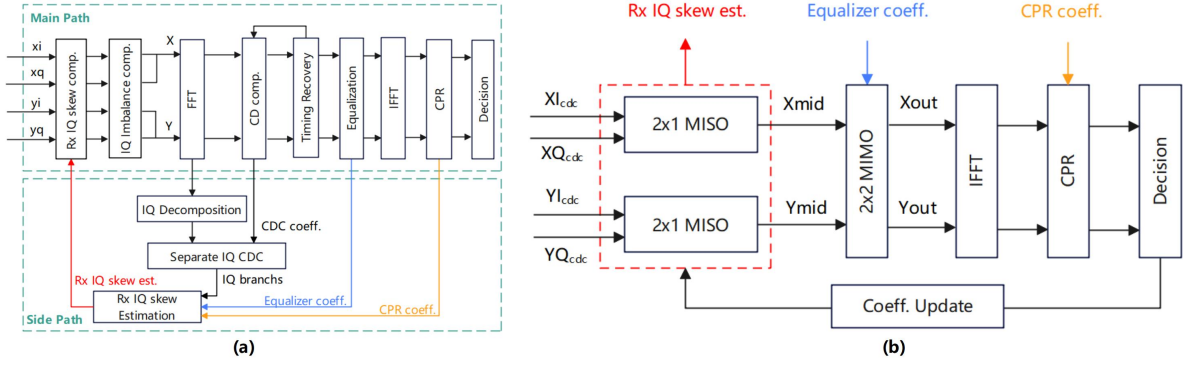


Fig. 1: (a) The proposed DSP scheme with a side path. (b) The principle of time-varying Rx IQ skew estimation.

X and Y polarizations, respectively, to compensate for the Rx IQ skew. The output signals of the 2×1 MISOs can be expressed as:

$$\begin{cases} X_{mid}(k) = XI_{cdc}(k) \cdot H_{XI}(k) + j \cdot XQ_{cdc}(k) \cdot H_{XQ}(k) \\ Y_{mid}(k) = YI_{cdc}(k) \cdot H_{YI}(k) + j \cdot YQ_{cdc}(k) \cdot H_{YQ}(k) \end{cases} \quad (2)$$

where XI_{cdc} , XQ_{cdc} , YI_{cdc} , YQ_{cdc} represent the IQ branches of X and Y polarizations after CDC, and H_{XI} , H_{XQ} , H_{YI} , H_{YQ} represent the 2×1 MISO equalizer coefficients. The output signals of the 2×2 MIMO can be expressed as:

$$\begin{cases} X_{out}(k) = X_{mid}(k) \cdot H_{XX}(k) + Y_{mid}(k) \cdot H_{XY}(k) \\ Y_{out}(k) = Y_{mid}(k) \cdot H_{YY}(k) + X_{mid}(k) \cdot H_{YX}(k) \end{cases} \quad (3)$$

where H_{XX} , H_{XY} , H_{YX} , H_{YY} represent the 2×2 MIMO equalizer coefficients. Since the architectures of the 2×2 MIMO and CPR modules are identical to those in the main-path DSP, the equalizer coefficients and CPR coefficients can be obtained from the main path, eliminating the need for recalculation. Therefore, only the equalizer coefficients of the 2×1 MISOs need to be updated. Based on the least mean squares (LMS) algorithm, the coefficients can be updated iteratively as:

$$\begin{cases} H_{XI}(k) = H_{XI}(k) + \mu \cdot e_x(k) \cdot (XI_{cdc}(k) \cdot H_{XX}(k))^* \\ H_{XQ}(k) = H_{XQ}(k) + \mu \cdot e_x(k) \cdot (XQ_{cdc}(k) \cdot H_{XX}(k))^* \cdot (-j) \\ H_{YI}(k) = H_{YI}(k) + \mu \cdot e_y(k) \cdot (YI_{cdc}(k) \cdot H_{YY}(k))^* \\ H_{YQ}(k) = H_{YQ}(k) + \mu \cdot e_y(k) \cdot (YQ_{cdc}(k) \cdot H_{YY}(k))^* \cdot (-j) \end{cases} \quad (4)$$

where μ denotes the step size for updating the equalizer coefficients, e_x and e_y represent the decision-directed errors of two polarizations in the frequency domain. Then, the Rx IQ skew estimates τ_x and τ_y can be estimated as:

$$\begin{cases} \tau_x = \partial \arg(H_{XI}(k) \cdot H_{XQ}^*(k)) / \partial \omega \\ \tau_y = \partial \arg(H_{YI}(k) \cdot H_{YQ}^*(k)) / \partial \omega \end{cases} \quad (5)$$

where ω is angular frequency, and $\arg(\cdot)$ represents argument of a complex number. Once the time-varying IQ skew is estimated, it can be compensated together with the fixed IQ skew using finite impulse response (FIR) fractional delay filters.

Since the main-path DSP tracks time-varying channel variations in real time, the side-path DSP should be activated only after the main-

path DSP has converged, allowing it to focus solely on compensating for time-varying Rx IQ skew. Given that the time-varying Rx IQ skew is typically within ± 0.5 ps and varies slowly on the order of seconds to minutes, it will not cause the main-path DSP to fail. Therefore, we can reduce the computational frequency of the estimation algorithm to track the skew slowly, thereby significantly reducing the power consumption of the side-path DSP while ensuring the stable convergence of the algorithm.

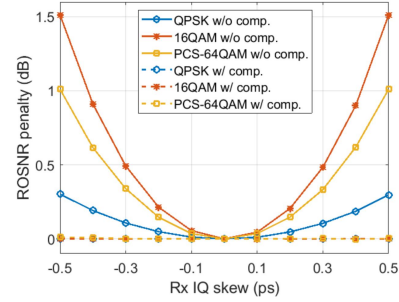


Fig. 2: ROSNR penalty vs. Rx IQ skew.

To validate our scheme through numerical simulation, we generate 250 GBaud signals using QPSK, 16QAM, and probabilistic constellation shaped 64QAM (PCS-64QAM). The Rx IQ skew is preset from -0.5 ps to 0.5 ps, with each simulation using a fixed value to approximate its slow variation over short intervals. The forward error correction (FEC) bit error rate (BER) threshold is $1.93\text{e-}2$ for QPSK and 16QAM, and $4.5\text{e-}2$ for PCS-64QAM. The ROSNR penalties versus Rx IQ skews are depicted in Fig. 2, with (w/) and without (w/o) Rx IQ skew estimation and compensation. It is shown that 0.5 ps Rx IQ skew may cause more than 1 dB ROSNR penalty for 16QAM and PCS-64QAM signals, while the ROSNR penalty becomes almost zero after time-varying Rx IQ skew estimation and compensation. To ensure that the ROSNR penalty of Rx IQ skew for next-generation 1.6 Tbps signals remains below 0.1 dB, the residual skew must be kept below approximately 0.1 ps, corresponding to 1/40 of the symbol period (denoted as T_s).

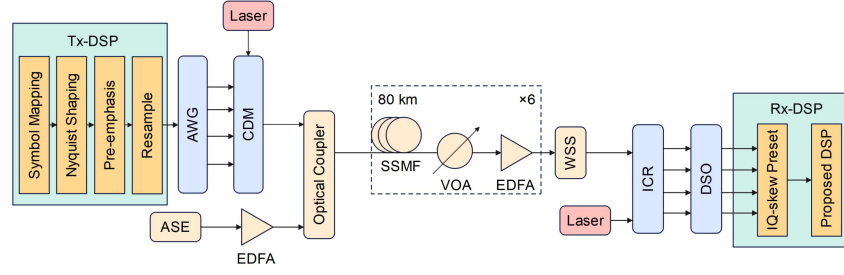


Fig. 3: The experimental setup.

Experimental setup

Fig. 3 shows the experimental setup of the transmission system. At the transmitter (Tx), 90 GBaud dual-polarization QPSK, 16QAM, and PCS-64QAM signals with a roll-off factor of 0.125 are pre-emphasised and resampled before being loaded into an arbitrary waveform generator (AWG) with a 65 GHz bandwidth and 224 GSa/s sampling rate. Two external cavity lasers at 193.6 THz serve as the optical carrier and local oscillator, respectively. The optical carrier is modulated by a coherent driver modulator (CDM) with a 58 GHz bandwidth. The amplified spontaneous emission (ASE) noise is amplified by an erbium-doped fiber amplifier (EDFA) and added to the optical signal by an optical coupler to adjust the OSNR. The transmission link comprises multiple standard single-mode fiber (SSMF) spans totaling 480 km in length, variable optical attenuators (VOAs), EDFAs, and a wavelength-selective switch (WSS). At the Rx, the optical signal is received by an integrated coherent receiver (ICR) with a 70 GHz bandwidth and sampled by a real-time digital storage oscilloscope (DSO) with a 59 GHz bandwidth. The Rx IQ skew is preset in Rx-DSP and the subsequent DSP chain is shown in Fig. 1(a). Since the experimental baudrate is lower than that of the simulation, we configure the Rx IQ skew as a proportion of T_s . The skew values for X and Y polarizations are equal in absolute value but opposite in sign.

Fig. 4(a) shows BER versus OSNR with and without time-varying Rx IQ skew estimation and compensation, where the Rx IQ skew is set to $0.108 T_s$. Fig. 4(b) shows the ROSNR penalty versus the Rx IQ skew. With Rx IQ skew estimation and compensation, the performance has been significantly enhanced, and the trend aligns with the simulation results. Fig. 4(c) shows the estimated value and error of Rx IQ skew for the 16QAM signal. When Rx IQ skew varies within $\pm 0.135 T_s$, the estimation error remains below $0.01 T_s$, without degrading the performance. We did not extend the study to larger skew values, as the current range is sufficient for time-varying skew estimation and

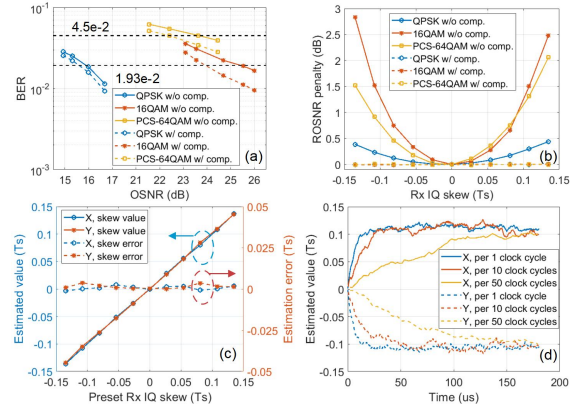


Fig. 4: (a) BER vs. OSNR (Rx IQ skew = $0.108 T_s$).

(b) ROSNR penalty vs. Rx IQ skew. (c) Estimated value and error of Rx IQ skew. (d) Convergence curve of Rx IQ skew estimation (Rx IQ skew = $0.108 T_s$).

compensation. Fig. 4(d) shows the convergence curve of Rx IQ skew estimation with different computational frequencies for the 16QAM signal, where the Rx IQ skew is set to $0.108 T_s$ and the number of symbols processed per clock cycle is set to 192. The proposed scheme achieves stable Rx IQ skew tracking within 200 μs , even when operating at a reduced computational frequency of one calculation per 50 clock cycles. Due to experimental constraints, longer sampling sequences cannot be acquired, but it can be inferred that the computational frequency can be further reduced.

Conclusions

We proposed an online low-power time-varying IQ skew estimation scheme for ultra-high baudrate coherent systems. The 90 GBaud experiment demonstrates that the proposed scheme can track Rx IQ skew of at least 13.5% of the symbol period within 200 μs , with an estimation error of less than 1% of the symbol period. This capability is sufficient for addressing time-varying Rx IQ skew. Additionally, the proposed scheme can flexibly reduce the algorithm's computational frequency to lower power consumption without compromising performance. The proposed scheme shows strong potential for estimating time-varying Rx IQ skew in next-generation optical transmission systems.

References

- [1] D. Li, Z. Chen, L. Zhang, P. Yi, J. Wang, H. Song, M. Cheng, S. Fu, M. Tang, Y. Qin, D. Liu and L. Deng, "Simultaneously Precise Calibration of Frequency Response and IQ Skew for 100Gbaud Optical Transceiver," presented at *2021 Optical Fiber Communications Conference and Exhibition (OFC)*, Washington, DC, USA, 2021, DOI: [10.1364/OFC.2021.Th5D.1](https://doi.org/10.1364/OFC.2021.Th5D.1).
- [2] R. Rios-Müller, J. Renaudier and G. Charlet, "Blind Receiver Skew Compensation and Estimation for Long-Haul Non-Dispersion Managed Systems Using Adaptive Equalizer," *Journal of Lightwave Technology*, vol. 33, no. 7, pp. 1315-1318, 2014, DOI: [10.1109/JLT.2014.2377582](https://doi.org/10.1109/JLT.2014.2377582).
- [3] Y. Fan, X. Su, H. Chen, J. Liang, Z. Tao, H. Nakashima and T. Hoshida, "Experimental Verification of IQ Imbalance Monitor for High-Order Modulated Transceivers," presented at *44th European Conference on Optical Communication (ECOC)*, Rome, Italy, 2018, DOI: [10.1109/ECOC.2018.8535555](https://doi.org/10.1109/ECOC.2018.8535555).
- [4] J. Liang, J. Chai, X. Chen, G. Lu, Z. Feng, L. Zhang and W. Wang, "Receiver IQ Imbalance and Skew Compensation for High Order Modulation Formats by Frequency Domain 4×2 MIMO," *Journal of Lightwave Technology*, vol. 41, no. 2, pp. 558-569, 2022, DOI: [10.1109/JLT.2022.3219866](https://doi.org/10.1109/JLT.2022.3219866).
- [5] N. Stojanovic and C. Xie, "An Efficient Method for Skew Estimation and Compensation in Coherent Receivers," *IEEE Photonics Technology Letters*, vol. 28, no. 4, pp. 489-492, 2015, DOI: [10.1109/LPT.2015.2500029](https://doi.org/10.1109/LPT.2015.2500029).
- [6] J. Liang, J. Chai, X. Chen and Q. Hu, "Timing Recovery Algorithms for Transceiver IQ skew Compensation and Monitoring," presented at *2022 Asia Communications and Photonics Conference (ACP)*, Shenzhen, China, 2022, DOI: [10.1109/ACP55869.2022.10088488](https://doi.org/10.1109/ACP55869.2022.10088488).
- [7] L. Fang, Y. Zhang, J. Chai, Z. Yan, T. Yang, J. Liang and X. Chen, "Anti-Singularity Clock Recovery and Rx/Tx IQ Skew Compensation Scheme," *IEEE Photonics Technology Letters*, vol. 35, no. 24, pp. 1359-1362, 2023, DOI: [10.1109/LPT.2023.3326837](https://doi.org/10.1109/LPT.2023.3326837).
- [8] W. Wang, Z. Wu, D. Zou, Q. Sui, C. Lu and F. Li, "Training Sequences Design for Simultaneously Transceiver IQ Skew Estimation in Coherent Systems," *Journal of Lightwave Technology*, vol. 42, no. 15, pp. 5088-5098, 2024, DOI: [10.1109/JLT.2024.3385422](https://doi.org/10.1109/JLT.2024.3385422).